# Homework 2 Solutions

**BSTA 551**

## Problem 1: Fisher Information and CRLB

A clinical trial is testing a new treatment for hypertension. The time (in days) until a patient achieves target blood pressure is modeled as an Exponential distribution with rate parameter $\lambda$ (where $E(X) = 1/\lambda$).

In a pilot study, the following times (in days) were observed for $n = 8$ patients:

$$X = \{12, 8, 15, 6, 10, 14, 9, 11\}$$

### Part (a)

**Calculate the Fisher Information $I(\lambda)$ for a single observation from an Exponential($\lambda$) distribution. Show your derivation using the definition $I(\lambda) = E[-\ell''(\lambda)]$.**

For $X \sim \text{Exponential}(\lambda)$, the pdf is $f(x; \lambda) = \lambda e^{-\lambda x}$ for $x > 0$.

**Log-likelihood for a single observation:**

$$\ell(\lambda) = \ln f(x; \lambda) = \ln \lambda - \lambda x$$

**First derivative:**

$$\ell'(\lambda) = \frac{1}{\lambda} - x$$

**Second derivative:**

$$\ell''(\lambda) = -\frac{1}{\lambda^2}$$

**Fisher Information:**

$$I(\lambda) = E[-\ell''(\lambda)] = E\left[\frac{1}{\lambda^2}\right] = \boxed{\frac{1}{\lambda^2}}$$

Note: Since the second derivative does not depend on $X$, the expectation is straightforward.

### Part (b)

**State the Cramér-Rao Lower Bound (CRLB) for unbiased estimators of $\lambda$ based on a sample of size $n$.**

For $n$ observations, the Fisher Information is:

$$I_n(\lambda) = n \cdot I(\lambda) = \frac{n}{\lambda^2}$$

The **Cramér-Rao Lower Bound** for unbiased estimators of $\lambda$ is:

$$\boxed{\text{CRLB} = \frac{1}{I_n(\lambda)} = \frac{\lambda^2}{n}}$$

### Part (c)

**The MLE of $\lambda$ is $\hat{\lambda} = 1/\bar{X}$. This estimator is biased. Find an unbiased estimator of $\lambda$ and compute its variance. Does this unbiased estimator achieve the CRLB?**

[JM NOTE: Apologies! I meant to give you the "hint" about the distribution of 1/Y used below and only realized I hadn't put it in the question when one of you asked me a question about it. Note that $\bar{X}$ is unbiased for $1/\lambda$ but $1/\bar{X}$ is not unbiased for $\lambda$ since $g(x) = 1/x$ is a convex function.]

For Exponential($\lambda$), we have $\sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$.

Using properties of the Gamma distribution, for $Y \sim \text{Gamma}(n, \lambda)$:

$$E\left(\frac{1}{Y}\right) = \frac{\lambda}{n-1} \quad \text{for } n > 1$$

Therefore:

$$E\left(\frac{1}{\bar{X}}\right) = E\left(\frac{n}{\sum X_i}\right) = \frac{n\lambda}{n-1}$$

So the MLE is biased with $E(\hat{\lambda}) = \frac{n}{n-1}\lambda > \lambda$.

**Unbiased estimator:**

$$\boxed{\hat{\lambda}_U = \frac{n-1}{n} \cdot \frac{1}{\bar{X}} = \frac{n-1}{\sum_{i=1}^n X_i}}$$

**Variance of the unbiased estimator:**

For $Y \sim \text{Gamma}(n, \lambda)$, we have:

$$\text{Var}\left(\frac{1}{Y}\right) = \frac{\lambda^2}{(n-1)^2(n-2)} \quad \text{for } n > 2$$

Therefore:

$$\text{Var}(\hat{\lambda}_U) = (n-1)^2 \cdot \text{Var}\left(\frac{1}{\sum X_i}\right) = (n-1)^2 \cdot \frac{\lambda^2}{(n-1)^2(n-2)} = \boxed{\frac{\lambda^2}{n-2}}$$

**Comparison with CRLB:**

$$\text{Var}(\hat{\lambda}_U) = \frac{\lambda^2}{n-2} > \frac{\lambda^2}{n} = \text{CRLB}$$

The unbiased estimator does **NOT** achieve the CRLB.

### Part (d)

**Using R, compute the MLE $\hat{\lambda}$ from the pilot study data. Also compute the CRLB at this estimated value. Interpret what this bound tells us about estimation precision.**

```r
# Pilot study data
bp_times <- c(12, 8, 15, 6, 10, 14, 9, 11)
n <- length(bp_times)

# MLE
lambda_mle <- 1 / mean(bp_times)

# Unbiased estimator
lambda_unbiased <- (n - 1) / sum(bp_times)

# CRLB at the MLE estimate
crlb <- lambda_mle^2 / n

cat("Sample size: n =", n, "\n")
```

```
Sample size: n = 8
```

```r
cat("Sample mean:", mean(bp_times), "days\n")
```

```
Sample mean: 10.625 days
```

```r
cat("MLE of  :", round(lambda_mle, 4), "per day\n")
```

```
MLE of  : 0.0941 per day
```

```r
cat("Unbiased estimate of  :", round(lambda_unbiased, 4), "per day\n")
```

```
Unbiased estimate of  : 0.0824 per day
```

```r
cat("CRLB at MLE:", round(crlb, 6), "\n")
```

```
CRLB at MLE: 0.001107
```

```r
cat("Standard error bound:", round(sqrt(crlb), 4), "\n")
```

```
Standard error bound: 0.0333
```

**Interpretation:** The CRLB tells us that no unbiased estimator of $\lambda$ can have variance smaller than 0.001107. This means the standard error of any unbiased estimator must be at least 0.0333. Our estimate of $\lambda \approx 0.0941$ per day (corresponding to a mean time of 10.6 days) has inherent uncertainty—we can expect our estimates to vary by roughly $\pm 0.067$ across repeated samples of this size.

## Problem 2: Method of Moments (Shifted Exponential Distribution)

The time (in hours) from hospital admission to discharge for patients undergoing a minor outpatient procedure can be modeled with a shifted (two-parameter) Exponential distribution. This accounts for a minimum required observation time $\theta$ plus additional random waiting time. The pdf is:

$$f(x; \lambda, \theta) = \lambda e^{-\lambda(x-\theta)}, \quad x \geq \theta, \quad \lambda > 0, \quad \theta > 0$$

For this distribution:

$$E(X) = \theta + \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

**Part (a)**

**Derive the Method of Moments estimators $\tilde{\lambda}$ and $\tilde{\theta}$ in terms of the sample mean $\bar{X}$ and sample standard deviation $S$.**

For the shifted Exponential distribution, the population moments are:

$$E(X) = \theta + \frac{1}{\lambda}, \quad E(X^2) = \text{Var}(X) + [E(X)]^2 = \frac{1}{\lambda^2} + \left(\theta + \frac{1}{\lambda}\right)^2$$

The corresponding sample moments are:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad \frac{1}{n}\sum_{i=1}^{n} X_i^2$$

**Method of Moments:** Set population moments equal to sample moments.

**First moment equation:**

$$E(X) = \bar{X}$$

$$\theta + \frac{1}{\lambda} = \bar{X} \qquad (1)$$

**Second moment equation:**

$$E(X^2) = \frac{1}{n}\sum_{i=1}^{n} X_i^2$$

It is easier to work with the variance. Recall that $\text{Var}(X) = E(X^2) - [E(X)]^2$, so:

$$E(X^2) - [E(X)]^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2$$

The left side is $\text{Var}(X) = \frac{1}{\lambda^2}$, and the right side is the sample variance (with $n$ in denominator):

$$\frac{1}{\lambda^2} = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \qquad (2)$$

Note that the sample standard deviation is $S = \sqrt{\frac{1}{n-1}\sum(X_i - \bar{X})^2}$, so:

$$\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{n-1}{n}S^2$$

Substituting into equation (2):

$$\frac{1}{\lambda^2} = \frac{n-1}{n} S^2 \qquad (3)$$

**Solving the system:**

**Step 1:** From equation (3), solve for $\lambda$:

$$\frac{1}{\lambda^2} = \frac{n-1}{n} S^2 \implies \lambda = \sqrt{\frac{n}{n-1}} \cdot \frac{1}{S}$$

**Step 2:** Substitute into equation (1) to find $\theta$:

$$\theta + \frac{1}{\lambda} = \bar{X}$$

$$\theta + S\sqrt{\frac{n-1}{n}} = \bar{X}$$

$$\theta = \bar{X} - S\sqrt{\frac{n-1}{n}}$$

**Method of Moments Estimators:**

$$\boxed{\tilde{\lambda} = \sqrt{\frac{n}{n-1}} \cdot \frac{1}{S} = \frac{1}{S}\sqrt{\frac{n}{n-1}}}$$

$$\boxed{\tilde{\theta} = \bar{X} - S\sqrt{\frac{n-1}{n}}}$$

### Part (b)

**Using R, calculate the MME estimates $\tilde{\lambda}$ and $\tilde{\theta}$. Verify that your estimate of $\theta$ is less than the minimum observed value.**

```
# Discharge times (hours)
discharge_times <- c(3.2, 4.1, 2.8, 5.3, 3.5, 4.8, 3.1, 6.2, 4.4, 3.9)
n <- length(discharge_times)

# Sample statistics
xbar <- mean(discharge_times)
s <- sd(discharge_times)  # This is S with (n-1) denominator

# Method of Moments estimates (exact formulas)
```

```r
lambda_mme <- (1/s) * sqrt(n / (n - 1))
theta_mme <- xbar - s * sqrt((n - 1) / n)

cat("Sample size: n =", n, "\n")
```

```
Sample size: n = 10
```

```r
cat("Sample mean:", round(xbar, 4), "hours\n")
```

```
Sample mean: 4.13 hours
```

```r
cat("Sample SD (S):", round(s, 4), "hours\n\n")
```

```
Sample SD (S): 1.0709 hours
```

```r
cat("MME estimates (exact formulas):\n")
```

```
MME estimates (exact formulas):
```

```r
cat("  ˜ =", round(lambda_mme, 4), "per hour\n")
```

```
  ˜ = 0.9843 per hour
```

```r
cat("  ˜ =", round(theta_mme, 4), "hours\n\n")
```

```
  ˜ = 3.1141 hours
```

```r
# Validity check
min_obs <- min(discharge_times)
cat("Validity check:\n")
```

```
Validity check:
```

```r
cat("  Minimum observed value:", min_obs, "hours\n")
```

```
  Minimum observed value: 2.8 hours
```

```
cat("  ~ =", round(theta_mme, 4), "hours\n")
```

```
 ~ = 3.1141 hours
```

```
cat("  Is ~ < min(X)?", theta_mme < min_obs, "\n\n")
```

```
  Is ~ < min(X)? FALSE
```

```
# Interpretation
cat("Interpretation:\n")
```

```
Interpretation:
```

```
cat("  Estimated minimum observation time:", round(theta_mme, 2), "hours\n")
```

```
  Estimated minimum observation time: 3.11 hours
```

```
cat("  Estimated mean additional wait time (1/ ):", round(1/lambda_mme, 2), "hours\n")
```

```
  Estimated mean additional wait time (1/ ): 1.02 hours
```

```
cat("  Estimated total mean discharge time:", round(theta_mme + 1/lambda_mme, 2), "hours\n")
```

```
  Estimated total mean discharge time: 4.13 hours
```

```
cat("  (Check: this should equal x̄ =", round(xbar, 2), "hours)\n")
```

```
  (Check: this should equal x̄ = 4.13 hours)
```

**Interpretation:** The MME estimates suggest that patients require a minimum of approximately 3.11 hours of observation, with an additional expected random waiting time of 1.02 hours before discharge. The validity check confirms that $\tilde{\theta}$ is less than the smallest observed discharge time (2.8 hours), which is necessary for the model to be consistent with the data.

## Problem 3: Maximum Likelihood Estimation (Devore 7.25)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pdf:

$$f(x;\theta) = (\theta+1)x^{\theta}, \quad 0 \le x \le 1, \quad \theta > -1$$

### Part (a)

**Derive the maximum likelihood estimator $\hat{\theta}$ of $\theta$.**

**Likelihood:**

$$L(\theta) = \prod_{i=1}^{n}(\theta+1)x_i^{\theta} = (\theta+1)^n \prod_{i=1}^{n} x_i^{\theta} = (\theta+1)^n \left(\prod_{i=1}^{n} x_i\right)^{\theta}$$

**Log-likelihood:**

$$\ell(\theta) = n\ln(\theta+1) + \theta\sum_{i=1}^{n}\ln(x_i)$$

**First derivative:**

$$\ell'(\theta) = \frac{n}{\theta+1} + \sum_{i=1}^{n}\ln(x_i)$$

**Setting $\ell'(\theta) = 0$:**

$$\frac{n}{\theta+1} = -\sum_{i=1}^{n}\ln(x_i)$$

$$\theta + 1 = \frac{n}{-\sum_{i=1}^{n}\ln(x_i)} = \frac{-n}{\sum_{i=1}^{n}\ln(x_i)}$$

$$\theta = \frac{-n}{\sum_{i=1}^{n}\ln(x_i)} - 1$$

**MLE:**

$$\boxed{\hat{\theta} = \frac{-n}{\sum_{i=1}^{n}\ln(x_i)} - 1 = -1 - \frac{1}{\overline{\ln(X)}}}$$

where $\overline{\ln(X)} = \frac{1}{n}\sum_{i=1}^{n}\ln(x_i)$.

**Verification (second derivative test):**

$$\ell''(\theta) = -\frac{n}{(\theta+1)^2} < 0$$

This confirms we have a maximum.

### Part (b)

**For a sample of size $n = 5$ with observations $x_1 = 0.92, x_2 = 0.79, x_3 = 0.85, x_4 = 0.95, x_5 = 0.88$, compute the MLE $\hat{\theta}$.**

```
x <- c(0.92, 0.79, 0.85, 0.95, 0.88)
n <- length(x)

# MLE calculation
sum_ln_x <- sum(log(x))
theta_mle <- -n / sum_ln_x - 1

cat("Observations:", x, "\n")
```

```
Observations: 0.92 0.79 0.85 0.95 0.88
```

```
cat("Sum of ln(x_i):", round(sum_ln_x, 4), "\n")
```

```
Sum of ln(x_i): -0.6607
```

```
cat("MLE ^ =", round(theta_mle, 4), "\n")
```

```
MLE ^ = 6.5672
```

### Part (c)

**Use the invariance property of MLEs to find the MLE of $\tau(\theta) = \frac{\theta+1}{\theta+2}$, which represents $E(X)$ for this distribution.**

By the **invariance property of MLEs**, the MLE of $\tau(\theta) = g(\theta)$ is $\hat{\tau} = g(\hat{\theta})$.

Therefore:

$$\hat{\tau} = \frac{\hat{\theta} + 1}{\hat{\theta} + 2}$$

```
tau_mle <- (theta_mle + 1) / (theta_mle + 2)
cat("MLE of  ( ) = ( +1)/( +2):", round(tau_mle, 4), "\n")
```

```
MLE of  ( ) = ( +1)/( +2): 0.8833
```

```r
cat("Sample mean (for comparison):", round(mean(x), 4), "\n")
```

```
Sample mean (for comparison): 0.878
```

Note: The MLE of $E(X)$ (0.8833) is close to but not exactly equal to the sample mean (0.878). This is expected since the MLE uses the full likelihood structure of the distribution.

## Problem 4: Sufficient Statistics (Medical Application)

A researcher is studying the number of adverse events reported per patient in a drug trial. The number of adverse events $X$ follows a Poisson($\mu$) distribution.

### Part (a)

**For a random sample $X_1, \dots, X_n$ from Poisson($\mu$), use the Neyman Factorization Theorem to show that $T = \sum_{i=1}^n X_i$ is a sufficient statistic for $\mu$.**

For $X_1, \dots, X_n \overset{iid}{\sim} \text{Poisson}(\mu)$, the joint pmf is:

$$f(x_1, \dots, x_n; \mu) = \prod_{i=1}^n \frac{e^{-\mu}\mu^{x_i}}{x_i!} = \frac{e^{-n\mu}\mu^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

We can factor this as:

$$f(x_1, \dots, x_n; \mu) = \underbrace{e^{-n\mu}\mu^{\sum x_i}}_{g(T;\mu)} \cdot \underbrace{\frac{1}{\prod_{i=1}^n x_i!}}_{h(x_1,\dots,x_n)}$$

where $T = \sum_{i=1}^n X_i$.

- $g(T;\mu) = e^{-n\mu}\mu^T$ depends on the data **only through** $T$ and involves $\mu$
- $h(x_1, \dots, x_n) = 1/\prod x_i!$ depends only on the data, **not on** $\mu$

By the **Neyman Factorization Theorem**, $T = \sum_{i=1}^n X_i$ is **sufficient for** $\mu$. $\square$

### Part (b)

**Show that the MLE $\hat{\mu} = \bar{X}$ is a function of the sufficient statistic $T$.**

The MLE is:

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{T}{n}$$

Since $\hat{\mu} = T/n$, the MLE can be written purely as a function of the sufficient statistic $T$. ✓

### Part (c)

In a clinical trial with $n = 50$ patients, suppose the following summary statistics are recorded:

$$\sum_{i=1}^{50} X_i = 72$$

A colleague suggests using a different estimator: $\tilde{\mu} = \frac{\text{number of patients with } X_i > 0}{n}$.

**Using R simulation, generate 1000 samples from Poisson$(\mu = 1.5)$ with $n = 50$ and compare the bias, variance and MSE of $\hat{\mu}$ and $\tilde{\mu}$.**

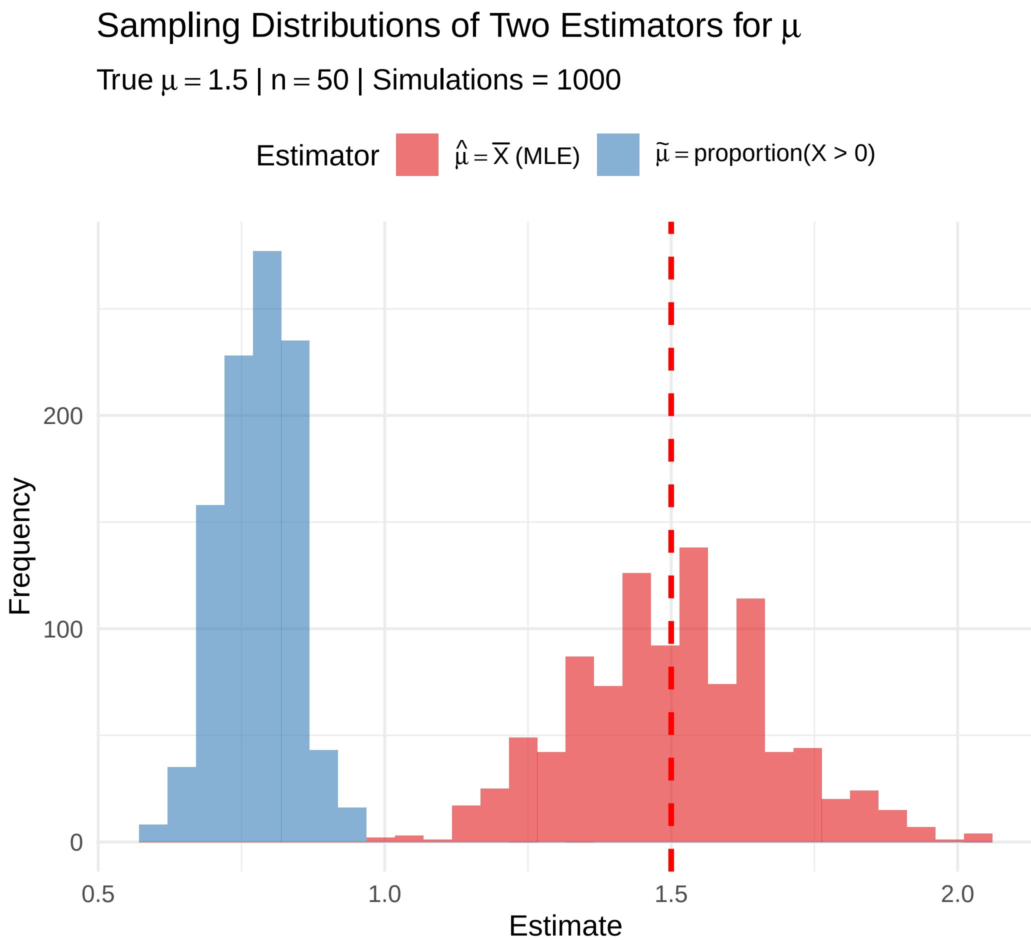
[JM side note: You don't need to use `map()` functions like I do, unless you want to learn them! Feel free to make this work however you know you to make tibbles (stack, make each vector separately, whatever).]

```r
set.seed(551)
n <- 50
mu_true <- 1.5
n_sims <- 1000

# Simulation
sim_results <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) rpois(n, mu_true)),
    mu_hat = map_dbl(data, mean),                    # MLE (based on sufficient stat)
    mu_tilde = map_dbl(data, \(x) mean(x > 0))       # Proportion with events
  )

# Compute summaries
summary_stats <- sim_results |>
  summarize(
    Estimator = c("ˆ = X̄ (MLE)", "˜ = proportion(X > 0)"),
```

```r
    True_Value = c(mu_true, 1 - exp(-mu_true)),  # Note: E(˜) = P(X > 0) = 1 - e^(- )
    Mean = c(mean(mu_hat), mean(mu_tilde)),
    Bias = c(mean(mu_hat) - mu_true, mean(mu_tilde) - mu_true),
    Variance = c(var(mu_hat), var(mu_tilde)),
    MSE = c(mean((mu_hat - mu_true)^2), mean((mu_tilde - mu_true)^2))
  )

print(summary_stats)
```

```
# A tibble: 2 x 6
  Estimator             True_Value  Mean     Bias Variance    MSE
  <chr>                      <dbl> <dbl>    <dbl>    <dbl>  <dbl>
1 ˆ = X̄ (MLE)                1.5   1.50   0.00384  0.0309  0.0309
2 ˜ = proportion(X > 0)      0.777 0.778 -0.722    0.00367 0.525
```

```r
# Visualization
sim_results |>
  pivot_longer(cols = c(mu_hat, mu_tilde),
               names_to = "Estimator", values_to = "Estimate") |>
  mutate(Estimator = factor(Estimator,
                            levels = c("mu_hat", "mu_tilde"),
                            labels = c("MLE", "Proportion"))) |>
  ggplot(aes(x = Estimate, fill = Estimator)) +
  geom_histogram(alpha = 0.6, bins = 30, position = "identity") +
  geom_vline(xintercept = mu_true, linetype = "dashed", color = "red", linewidth = 1) +
  labs(title = bquote("Sampling Distributions of Two Estimators for" ~ mu),
       subtitle = bquote("True" ~ mu == .(mu_true) ~ "|" ~ n == .(n) ~ "| Simulations =" ~ .
       x = "Estimate", y = "Frequency") +
  scale_fill_brewer(palette = "Set1",
                    labels = c(expression(hat(mu) == bar(X) ~ "(MLE)"),
                               expression(tilde(mu) == "proportion(X > 0)"))) +
  theme_minimal() +
  theme(legend.position = "top")
```




**Interpretation:**

- The MLE $\hat{\mu} = \bar{X}$ is essentially unbiased with bias 0 and estimates $\mu$ directly.
- The proportion estimator $\tilde{\mu}$ is heavily biased because it estimates $P(X > 0) = 1 - e^{-\mu} \approx 0.777$, not $\mu$ itself.
- Even though $\tilde{\mu}$ has smaller variance, its large bias results in much larger MSE.
- The MLE, which is based on the sufficient statistic $T = \sum X_i$, is clearly superior for estimating $\mu$.

### Problem 5: MLE for Normal Variance (Devore 7.31 Modified)

Measurement errors in a laboratory instrument are assumed to follow a Normal distribution with known mean $\mu = 0$ and unknown variance $\sigma^2$. A sample of $n$ measurements yields observations $X_1, \ldots, X_n$.

### Part (a)

**Derive the MLE of $\sigma^2$.**

For $X_i \overset{iid}{\sim} N(0, \sigma^2)$, the pdf is:

$$f(x; \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)}$$

**Likelihood:**

$$L(\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x_i^2/(2\sigma^2)} = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_{i=1}^n x_i^2}{2\sigma^2}\right)$$

**Log-likelihood:**

$$\ell(\sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{\sum_{i=1}^n x_i^2}{2\sigma^2}$$

**Derivative with respect to $\sigma^2$:**

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n x_i^2}{2(\sigma^2)^2}$$

**Setting equal to zero:**

$$\frac{n}{2\sigma^2} = \frac{\sum_{i=1}^n x_i^2}{2(\sigma^2)^2}$$

$$n\sigma^2 = \sum_{i=1}^{n} x_i^2$$

**MLE:**

$$\boxed{\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2}$$

### Part (b)

**Is this MLE unbiased? If not, find the bias.**

Since $X_i \sim N(0, \sigma^2)$, we have:

$$E(X_i^2) = \text{Var}(X_i) + [E(X_i)]^2 = \sigma^2 + 0 = \sigma^2$$

Therefore:

$$E(\hat{\sigma}^2) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) = \frac{1}{n} \cdot n \cdot \sigma^2 = \sigma^2$$

The MLE **is unbiased**:

$$\boxed{\text{Bias}(\hat{\sigma}^2) = E(\hat{\sigma}^2) - \sigma^2 = 0}$$

### Part (c)

**A technician takes $n = 15$ measurements and records: $\sum_{i=1}^{15} X_i^2 = 4.8$**

**Calculate the MLE of $\sigma^2$ and provide an unbiased estimate.**

```r
n <- 15
sum_x_squared <- 4.8

# MLE (which is also unbiased in this case)
sigma2_mle <- sum_x_squared / n

cat("n =", n, "\n")
```

```
n = 15
```

```r
cat("Sum of X_i² =", sum_x_squared, "\n\n")
```

```
Sum of X_i² = 4.8
```

```r
cat("MLE of  ²:", round(sigma2_mle, 4), "\n")
```

```
MLE of  ²: 0.32
```

```r
cat("Unbiased estimate of  ²:", round(sigma2_mle, 4), "(same as MLE)\n")
```

```
Unbiased estimate of  ²: 0.32 (same as MLE)
```

```r
cat("Estimated  :", round(sqrt(sigma2_mle), 4), "\n")
```

```
Estimated  : 0.5657
```

**Note:** In this special case where $\mu = 0$ is known, the MLE $\hat{\sigma}^2 = \frac{1}{n}\sum X_i^2$ is already unbiased. This is different from the usual case where $\mu$ is unknown, in which case the MLE $\frac{1}{n}\sum(X_i - \bar{X})^2$ is biased and we use $\frac{1}{n-1}\sum(X_i - \bar{X})^2$ for an unbiased estimate.